

Exact Traces and Escape Ranks for a Three-Symbol Open Walsh Gate

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Abstract

We study a finite three-symbol Walsh gate obtained by passing one qutrit per step through a rank-two Fourier-projection contraction. The gate has exact left and right defect projections, a tensor-power identity after one full factor cycle, and an all-period trace formula governed by a greatest common divisor. Newton recursion then gives exact secular polynomials. A tempting one-step rank formula is false: the rank is $2 \cdot 3^{k-1}$, while 2^k occurs only after k steps. The construction is a finite scattering subgate with a closed unitary parent, not a self-adjoint or target-matched quantization.

1 Frozen open gate

Let $\omega = e^{2\pi i/3} = (-1 + i\sqrt{3})/2$ and

$$(F_3)_{j\ell} = 3^{-1/2} \omega^{j\ell}, \quad P = \text{diag}(1, 0, 1), \quad A = F_3^* P.$$

On $\mathcal{H}_k = (\mathbb{C}^3)^{\otimes k}$, with lexicographic qutrit basis, freeze

$$B_k(v_0 \otimes \cdots \otimes v_{k-1}) = v_1 \otimes \cdots \otimes v_{k-1} \otimes Av_0. \quad (1)$$

One application is one clock tick, and $D_k(z) = \det(I_{3^k} - zB_k)$. No spectral rescaling is performed.

Theorem 1. *For every $k, n \geq 1$, with $d = \gcd(n, k)$,*

$$\|B_k\| = 1, \quad \text{rank } B_k = 2 \cdot 3^{k-1}, \quad B_k^k = A^{\otimes k}, \quad \text{rank } B_k^k = 2^k, \quad (2)$$

$$\text{Tr}(B_k^n) = \text{Tr}(A^{n/d})^d. \quad (3)$$

The subunitarity defects are the rank- 3^{k-1} projections

$$I - B_k^* B_k = (I - P) \otimes I_{3^{k-1}}, \quad (4)$$

$$I - B_k B_k^* = I_{3^{k-1}} \otimes F_3^* (I - P) F_3. \quad (5)$$

Proof. Write $S(v_0, \dots, v_{k-1}) = (v_1, \dots, v_{k-1}, v_0)$ and $C = I^{\otimes(k-1)} \otimes A$. Then $B_k = CS$, while $A^* A = P$ and $AA^* = F_3^* P F_3$. This proves the norm, one-step rank, and both Gram formulas. Tracking a pure tensor for k steps sends every factor through A once and restores its position, proving the tensor identity and its rank.

For (3), multiplication of basis matrix elements identifies factor positions along the cycles of addition by n modulo k . This permutation has d cycles, indexed by residues modulo d . During the n successive shifts, the departing positions are $0, 1, \dots, n-1$ modulo k ; each residue modulo d occurs exactly n/d times. Thus every permutation cycle contracts exactly n/d ordered copies of A , and its free qutrit-index sum is $\text{Tr}(A^{n/d})$. The d cycles are independent, giving the displayed power. This proves every period; no finite replay is used. $\square$

The distinction in (2) is substantive. One tick opens only the factor crossing the cut; the full tensor opening occurs only after k ticks.

2 Secular and path formulas

The one-qutrit characteristic polynomial is

$$\det(\lambda I - A) = \lambda(\lambda^2 - \tau\lambda + q), \quad \tau = \frac{\sqrt{3}}{6} - \frac{i}{2}, \quad q = -\frac{1}{2} - \frac{\sqrt{3}i}{6}.$$

Put $t_0 = 2, t_1 = \tau$ and $t_m = \tau t_{m-1} - q t_{m-2}$. Thus $t_m = \text{Tr}(A^m)$. If $D_k(z) = \sum_j c_{k,j} z^j$, then

$$c_{k,0} = 1, \quad c_{k,m} = -\frac{1}{m} \sum_{j=1}^m c_{k,m-j} t_{j/\gcd(j,k)}^{\gcd(j,k)}. \quad (6)$$

This is a reproducible expression in $\mathbb{Q}(\sqrt{3}, i)$. In particular,

$$\begin{aligned} D_1(z) &= 1 + \left(-\frac{\sqrt{3}}{6} + \frac{i}{2}\right) z + \left(-\frac{1}{2} - \frac{\sqrt{3}i}{6}\right) z^2, \\ D_2(z) &= 1 + \left(-\frac{\sqrt{3}}{6} + \frac{i}{2}\right) z + \left(-\frac{\sqrt{3}}{6} + \frac{i}{6}\right) z^3 + \left(-\frac{1}{6} - \frac{\sqrt{3}i}{6}\right) z^4. \end{aligned}$$

To prove the endpoint, triangularize A . It has one zero and two nonzero eigenvalues, counted algebraically, so $A^{\otimes k} = B_k^k$ has exactly $3^k - 2^k$ zero eigenvalues. Spectral mapping preserves that total at zero: zero is the only eigenvalue of B_k whose k th power is zero. Hence

$$\det(\lambda I - B_k) = \lambda^{3^k - 2^k} \sum_{j=0}^{2^k} c_{k,j} \lambda^{2^k - j}, \quad (7)$$

and the last coefficient is nonzero. The exact payload lists every coefficient through $k = 5$; its compact receipt is

k	3^k	rank B_k	deg D_k	nonzero $c_{k,j}$
1	3	2	2	3
2	9	6	4	4
3	27	18	8	9
4	81	54	16	13
5	243	162	32	21

In the frozen basis, a directed edge carries its complex matrix entry. Matrix multiplication and primitive-root decomposition give

$$\text{Tr}(B_k^n) = \sum_{w \text{ rooted closed}} a(w), \quad D_k(z) = \prod_{[p] \text{ primitive}} (1 - a(p) z^{|p|}). \quad (8)$$

All signed and complex cancellations are retained. Since the maximum absolute column sum is $\sqrt{3}$, the absolute rooted-path mass at time n is at most $3^k (\sqrt{3})^n$. Thus the logarithmic majorant converges for $|z| < 1/\sqrt{3}$, where the raw product is absolutely regroupable. The determinant is globally a polynomial; no larger raw product domain is claimed.

3 Controls and boundary

For $P = I_3$, $A = F_3^*$ and the corresponding B_k is unitary of rank 3^k , with vanishing defects. This is the closed parent from which the opening is defined. Projector order is an isospectral negative control: if $A_R = P F_3^*$, then

$$A_R = F_3 A F_3^*, \quad \text{Tr}(A_R^m) = \text{Tr}(A^m).$$

Formula (3) therefore leaves every D_k unchanged, although matrix entries and the left/right Gram placements change. By contrast, moving the hole to $P_0 = \text{diag}(0, 1, 1)$ preserves opening rank but gives

$$\text{Tr}(F_3^* P_0) = -\frac{\sqrt{3}}{3} - i, \quad [z] D_k^{(P_0)} = \frac{\sqrt{3}}{3} + i,$$

rather than $[z] D_k = -\sqrt{3}/6 + i/2$. Hole location, but not this projector order, changes the secular data.

Exact replay checks 60 traces and all 67 coefficients for $k = 1, \dots, 5$. The $k = 2$ prefix through period eight totals 510 rooted paths and 71 primitive cycles; it is not the proof of (2), (3), or (8). A producer-independent checker passes 748 assertions, SymPy passes 141 checks, byte replay passes, and 40 repaired-hash plus one stale-hash mutations are rejected.

The strict Route-A verdict is

$$\text{A1_WEAK} / \text{A2_FAIL} / \text{A3_FAIL} / \text{A4_UNITARY_OR_SCATTERING_CANDIDATE}, \\ \text{overall ROUTE_A_EXPLORATORY}.$$

This finite- k subunitary gate is not self-adjoint and has no semiclassical target matching. We claim no target divisor or counting law, prime-like map, arithmetic local data, Euler factor, root number, automorphy, antiunitary symmetry, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.